Faculty of Informatics Engineering

Department of Software Engineering



SMART RESTAURANT

Senior 2 Graduation Project Report — Part 1

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Executive Project Summary

The 'Integrated Smart Restaurant System' aims to develop a comprehensive web platform that unifies core restaurant operations into a single digital environment. The system allows customers to explore restaurants and menus, make reservations, place food orders, execute payments, track delivery status, receive AI-driven meal recommendations, interact with a conversational assistant, and access surplus food donation workflows. The baseline system built in Senior 1 utilized React for the frontend, Django REST Framework for backend web services, and MySQL for relational data persistence.

The Senior 2 phase introduces two major system expansions: first, an end-to-end food delivery workflow tracking orders from customer address specification to real-time fulfillment updates; and second, an image-based nutritional estimation pipeline leveraging computer vision models to identify meal items from photos and map them to nutritional databases. Because visual estimation cannot perfectly account for hidden ingredients or exact weight, results are presented as approximate estimates with explicit confidence boundaries and user portion confirmation.

1. Scientific Problem & Project Objectives

1.1 Project Background

Traditional restaurant digital services are often fragmented across disparate platforms: one for table booking, another for delivery fulfillment, and separate tools for food waste reduction. This operational fragmentation causes redundant data entry, inconsistent order tracking, and increased operational overhead. The proposed solution unifies these functional verticals within a cohesive, role-based architecture.

1.2 Scientific Problem Formulation

The core scientific challenge lies in architecting a unified software ecosystem that seamlessly synchronizes restaurant service cycles with delivery logistics while simultaneously providing visual nutritional estimations from dish photographs—all without compromising data integrity, user security, or system responsiveness. The problem breaks down into three key pillars:

- **Operational Integration:** How to maintain consistent state transitions across restaurant managers, customers, delivery riders, and charity entities within a unified relational schema.
- **Visual Content Estimation:** How to extract candidate food classes from image uploads, estimate portion size, and map identified classes to nutrient data while acknowledging inherent estimation error.
- **Reliability & User Experience:** How to enforce strict role-based access control, secure user uploaded media, and ensure resilient error handling during third-party service degradation.

1.3 Primary Objective

To design, expand, and evaluate a smart web-based restaurant management platform that delivers unified ordering, reservation, and delivery management alongside an automated image-based calorie estimation feature, built on a modular, maintainable, and fully testable architecture.

1.4 Specific Sub-Objectives

- **Delivery Architecture:** Define lifecycle states for delivery orders, from customer creation to final doorstep fulfillment.
- **Address & Fee Logic:** Implement address selection, distance/fee calculations, and real-time status updates.
- **Managerial Control:** Enable restaurant managers to approve orders, update preparation status, and assign dispatch riders.
- **Computer Vision Pipeline:** Structure an image processing pipeline for user dish uploads with quality validation.
- **Nutritional Mapping:** Connect identified food items to reference nutritional databases (e.g., USDA FoodData Central).
- **Portion Adjustment:** Provide portion sizing controls and confidence disclaimers to ensure transparent estimations.
- **Baseline Preservation:** Maintain full backward compatibility with Senior 1 baseline features (React, Django, MySQL, AI Chat, Recommendations, and Charity workflows).

1.5 Senior 2 Project Scope

Domain Vertical	Proposed Scope Details	Scope Status
Core Platform	Accounts, Restaurants, Menus, Reservations, Orders, Payments, AI Chat, Recommendations, Analytics, Donations.	Preserved from Senior 1 Baseline
Delivery Module	Delivery addresses, estimated fee/time calculation, multi-stage state machine, rider assignment, tracking.	New Feature (Senior 2)
Vision-Based Nutrition	Image upload, food recognition, portion size confirmation, USDA data query, calorie calculation.	New Feature (Senior 2)

2. State of the Art & Related Systems Review

A rigorous comparative evaluation was performed against existing commercial platforms and academic benchmarks to pinpoint structural functional gaps:

- **OpenTable & Resy:** Excellence in table booking and availability tracking; lack integrated online ordering, delivery logistics, and visual nutritional analysis.
- **GloriaFood:** Robust digital menu ordering and restaurant workflows; lacks advanced analytics and AI capabilities.
- **Toast POS:** Complete point-of-sale operational suite; lacks native conversational AI, visual calorie estimation, and surplus food donation features.
- **Too Good To Go:** Industry leader in food waste reduction and surplus food sales; does not provide full-service dining management or AI features.
- **Food-101 & Nutrition5k Benchmarks:** Academic datasets demonstrating food image classification and deep-learning visual mass/nutrient prediction; emphasize the necessity of user portion verification due to hidden ingredient uncertainty.

3. Proposed Solution Architecture

The proposed system adopts a decoupled client-server architecture consisting of a React Single Page Application (SPA), RESTful Django controllers, modular business logic services, and a MySQL relational database.

3.1 Delivery Fulfillment Workflow

The delivery lifecycle follows a rigorous state machine: Customer Order Creation -> Address & Fee Verification -> Restaurant Acceptance & Preparation -> Rider Assignment -> Dispatch & Tracking -> Final Fulfillment Confirmation.

3.2 Image-Based Calorie Calculation Logic

Nutritional estimation is computed using the standard mass-energy formulation:

$$\text{Estimated Calories (kcal)} = \left(\text{Portion Mass (g)} / 100 \right) \times \text{Energy per 100g (kcal)}$$

When multi-item dishes are detected, individual component energies are calculated independently and summed, providing a granular itemized breakdown.

Chapter 1: Theoretical Study & Technological Foundations

1.1 Layered Architectural Pattern

The system enforces strict separation of concerns across five primary tiers: Presentation Layer (React), API Controller Layer (Django REST), Business Logic Layer (Domain Services), Data Access Layer (Django ORM), and Database Persistence Layer (MySQL).

1.2 Core Technological Stack

- **React Frontend:** Component-driven architecture providing dynamic interfaces, centralized state handling, and responsive user feedback.
- **Django REST Framework:** Provides robust API endpoint routing, request validation, authentication filters, and serialization routines.
- **MySQL Database:** Relational data store enforcing transactional consistency, relational constraints, and indexed queries across domain entities.
- **Computer Vision Services:** Deep learning food classification model integrated via RESTful backend service handlers.

Conclusion & Expected Deliverables

This report establishes the baseline requirements, system architecture, and theoretical foundations for the Senior 2 expansion of the Smart Restaurant System. The addition of food delivery logistics and computer-vision nutritional estimation significantly enhances the platform's utility for both commercial restaurant operators and wellness-conscious consumers.

References

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